

ABERDARE TECHNICAL COLLEGE

ICT TECHNICIAN LEVEL 5

CONTINUOUS PRACTICAL –PRACTICAL 2

UNIT: NETWORK DESIGN AND MANAGEMENT

ISCED UNIT CODE: 0612 451 07A

ELEMENT COVERED: INSTALL COMPUTER NETWORK

TIME: 2 Hours

TOTAL MARKS: 50 Marks

Student Name:

Admno Number:

Date:.....

PRACTICAL TASK: COMPUTER NETWORK SETUP AND CABLING

Instructions to the Candidate

You are required to construct a network cable, set up a Local Area Network (LAN) consisting of four computers, a switch, and a wireless router, configure network devices, and verify connectivity. Read all instructions carefully before starting the practical exercise.

Materials and Equipment Provided

- UTP Cat6 cable
- RJ45 connectors
- Cable crimper
- Cable stripper

- Network cable tester
- Network switch
- Wireless router
- Four computers
- Patch cords
- Cable ties
- Screwdrivers and other networking tools

TASK 1: Network Cable Construction (20 Marks)

Construct one straight-through Ethernet cable.

Instructions

1. Measure and cut 1meter of UTP cable.
2. Strip the outer jacket of the cable carefully using a cable stripper.
3. Arrange the conductors according to the appropriate cabling standard.
4. Insert the conductors fully into the RJ45 connector.
5. Crimp the connector using a cable crimper.
6. Repeat the termination process on the opposite end of the cable.
7. Test the cable using a network cable tester.
8. Present the completed cable to the assessor for verification.

TASK 2: Network Installation and Setup (20 Marks)

Using the provided networking devices, set up the network according to the given design.

Instructions

1. Connect all four computers to the switch using network cables.

2. Connect the switch to one of the LAN ports of the wireless router.
3. Power on all devices.
4. Organize and secure all cables using cable ties.
5. Configure the router using the details provided below:

Setting	Value
Router IP Address	192.168.1.1
Subnet Mask	255.255.255.0
Wireless Network Name (SSID)	ATC-NETWORK
Wireless Password	Network@123

6. Configure the following IP addresses on the computers:

Device	IP Address
PC 1	192.168.1.10
PC 2	192.168.1.11
PC 3	192.168.1.12
PC 4	192.168.1.13

7. Verify communication between computers using the **Ping** command.
8. Verify communication between each computer and the router.
9. Demonstrate successful connectivity to the assessor.

TASK 3: Network Testing and Documentation (10 Marks)

Test the network and prepare a brief report.

Instructions

Prepare a report containing:

1. Title of the practical activity.
2. List of networking tools used.
3. List of networking materials used.
4. Cabling standard used.
5. Router configuration details.
6. IP addresses assigned to computers.
7. Results of cable testing.
8. Results of connectivity testing.
9. Challenges encountered during setup.
10. Recommendations for improving the network.